

REQUIREMENT SPECIFICATION DOCUMENT E-COMMERCE MANAGEMENT SYSTEM (EMS) ONLINE TOY STOR

1. INTRODUCTION :

This section introduces the Online Toy Store as a digital shopping platform that supports online toy purchasing. It explains that the system helps manage toys, customers, orders, payments, and deliveries, making shopping easy, convenient, and accessible for parents and children anytime and anywhere.

2. SCOPE :

Here we define what the Online Toy Store will achieve and why it's useful:

- Customers can browse and purchase toys online.
- Sellers can upload and manage toy products easily.
- Orders and payments are processed digitally.
- Customers can track their orders.
- The system can grow with the increasing number of customers and products (scalable).

3. PROBLEM STATEMENT :

This explains the issues in traditional toy shopping:

- Customers have limited choices in nearby toy stores.
- Visiting multiple stores takes time and effort.
- Manual inventory and order management can lead to errors.
- Customers cannot easily compare toy products and prices.

The Online Toy Store solves these problems by digitizing and automating the shopping and management processes.

4. STAKEHOLDERS :

Lists the people who interact with the system:

- Customers (browse toys, place orders, and track deliveries).
- Sellers (upload products, update stock, and manage orders).
- Administrators (manage users, products, and reports).
- Delivery Staff (deliver products and update delivery status).
- Payment Gateway (process online payments).

5. FUNCTIONAL REQUIREMENTS

These are the core tasks the system must perform.

- User Registration and Login: Customers can create an account by entering their personal details and log in securely using their email or username and password.
- Product Browsing and Search: Customers can browse toys by category, age group, brand, or price. They can also search for specific toys using the search feature.

- **Shopping Cart Management:** Customers can add toys to the shopping cart, update quantities, remove unwanted items, and view the total price before placing an order.
- **Order Placement:** Customers can place orders by selecting toys from the shopping cart, entering the delivery address, choosing a payment method, and confirming the purchase.
- **Online Payment:** The system allows customers to make secure online payments through UPI, debit card, credit card, net banking, or cash on delivery.
- **Order Tracking:** Customers can track the status of their orders such as Ordered, Packed, Shipped, Out for Delivery, and Delivered. They can also view their order history.
- **Product and Inventory Management:** The administrator can add new toys, update product details, remove unavailable products, manage categories, and maintain stock availability.
- **Customer Management:** The administrator can manage customer accounts, update customer information, handle customer queries, and monitor purchase history.
- **Report Generation:** The system generates reports such as sales reports, order reports, inventory reports, and customer reports to help the administrator monitor business performance.

Think of this as the "must-have features" of the Online Toy Store.

6. NON-FUNCTIONAL REQUIREMENTS :

These describe the quality attributes of the system.

- **Performance:** The system should load pages quickly, process customer requests without delay, and support multiple users at the same time.
- **Security:** Customer accounts, passwords, and payment information should be securely protected. Only authorized administrators should be allowed to modify product and order details.
- **Reliability:** The system should operate continuously with minimum downtime and maintain accurate customer, product, and order information.
- **Usability:** The interface should be simple, attractive, and easy to navigate so that customers can purchase toys without difficulty.
- **Scalability:** The system should be able to support an increasing number of customers, products, and transactions as the business grows.
- **Availability:** The Online Toy Store should be available 24 hours a day, allowing customers to shop anytime from any location.
- **Compatibility:** The website should work smoothly on desktops, laptops, tablets, and smartphones using different web browsers.
- **Maintainability:** The system should be easy to maintain, update, fix errors, and add new features whenever required. These ensure the Online Toy Store is reliable, safe, and user-friendly.

7. SYSTEM MODULES :

Breaks the Online Toy Store into functional components:

- **Customer Management:** Register customers, manage profiles, addresses, and purchase history.
- **Product Management:** Add, edit, delete, and organize toys with details such as toy name, brand, category, age group, price, description, and stock.
- **Category Management:** Organize toys into categories such as Educational Toys, Soft Toys, Action Figures, Board Games, Building Blocks, Remote Control Toys, and Outdoor Toys.

- Shopping Cart Management: Allow customers to add toys, update quantities, remove items, and calculate the total purchase amount.
- Order Management: Process customer orders, update order status, maintain order history, and manage cancellations and returns.
- Payment Management: Handle online payment transactions, record payment details, and generate payment receipts.
- Inventory Management: Maintain toy stock, update available quantities after every purchase, and notify the administrator when stock is low.
- Delivery Management: Track shipping details, update delivery status, and confirm successful delivery of customer orders.
- Report Management: Generate sales reports, inventory reports, customer reports, and order reports to help the administrator analyze business performance.

Each module focuses on a specific area of the system.

8. INPUT REQUIREMENTS :

Defines the data the system needs to work:

- Customer information (profile and address).
- Toy details (name, brand, category, age group, price).
- Login credentials.
- Order details.
- Payment information.

Inputs are the raw data entered into the Online Toy Store.

9. OUTPUT REQUIREMENTS :

Specifies the results the system produces:

- Customer account confirmation.
- Product search results.
- Order confirmation.
- Payment receipt.
- Order tracking details.
- Sales and inventory reports.

Outputs are what users see after the system processes inputs.

10. CONSTRAINTS :

Lists the boundaries and challenges:

- Requires stable internet connectivity.
- Real payment gateway integration may not be included.
- Live delivery tracking may not be supported.
- Must comply with customer data privacy regulations.

These are the practical limits the system must work within.

CONCLUSION :

Summarizes that the Online Toy Store is a secure, scalable solution that modernizes toy shopping, making it more accessible and efficient for both customers and businesses.